

Q.35 Define Lami's Theorem.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

Q.36 Define ultimate stress, working stress and factor of safety. What is the relationship between them?

Q.37 Explain the method of resolution of finding the resultant of a number of coplanar concurrent forces.

Q.38 A cantilever beam of span 5m carries a uniformly distributed load of 4kN per m over the whole span. Draw its shear force and bending moment diagram

No. of Printed Pages : 4

180245/120245/030245

Roll No.

4th Semester/ Arch
Subject: Structural Mechanics

Time : 3 Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Applied mechanics deals with the
a) Laws of mechanics b) Laws of chemistry
c) Laws of physics d) Laws of forces
- Q.2 Density is a _____ quality.
a) Vector b) Scalar
c) Both A & B d) None of the above
- Q.3 The amount of matter contained in a body is called its
a) Mass b) Weight
c) Volume d) Density
- Q.4 Lami's Theorem can be applied when the body is
a) Vibrating b) Moving
c) Rotating d) In equilibrium
- Q.5 The deformation per unit length is called
a) Strain b) Stress
c) Elasticity d) Shear

- Q.6 The ability of a material to deform without breaking is called
- a) Elasticity b) Plasticity
c) Creep d) Ductility
- Q.7 The unit of strains is
- a) cm/cm b) m/m
c) N/cm² d) No unit
- Q.8 The point of contra flexure occurs in case of
- a) Cantilever
b) Simply supported beam
c) Overhanging beam
d) Both A & B
- Q.9 The rate of change of bending moment is equal to
- a) Shear force b) Slope
c) Deflection d) None of these
- Q.10 Which of the following is a dimension less quantity
- a) Shear stress b) Poisson's ratio
c) Strain d) Both A & C

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (10x1=10)

- Q.11 In SI System, the unit of mass is ____.
- Q.12 Force is a _____ quantity.
- Q.13 Coplaner forces lies in _____ plane.
- Q.14 Definition of forces follows from Newton's _____ law of motion.

(2) 180245/120245/030245

- Q.15 The unit of stress is ____.
- Q.16 The effect of tensile force is to _____ the length of the body.
- Q.17 The positive bending moment is called ____ moment.
- Q.18 A beam whose _____ are fixed is known as fixed beam.
- Q.19 IN a cantilevel beam maximum B.M. occurs at the _____ support.
- Q.20 The unit of moment of inertia is ____.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Define force with their four characteristics.
- Q.22 Write a short note one Equilibrium of Rigid bodies.
- Q.23 State principle of transmissibility of forces.
- Q.24 Explain parallelogram law of forces.
- Q.25 Describe stress and types of stresses.
- Q.26 State polygon law of forces.
- Q.27 Describe elasticity and elastic limit.
- Q.28 Explain bulk modulus of elasticity.
- Q.29 Differentiate between shear force and bending moment. (any four) with diagram.
- Q.30 Explain moment of inertia.
- Q.31 Define Varignon's Theorem.
- Q.32 Describe pin jointed frames.
- Q.33 Differentiate between point load and U.D.L. (Any five)
- Q.34 Explain point of contraflexure.

(3) 180245/120245/030245